

UTA Smart Kiosk

Team Name

The Kiosk Guys

Timeline

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Students

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Abstract

The Smart Kiosk is an interactive campus system developed to improve how students, faculty, staff, and visitors engage with UTA's Engineering Research South A (ERSA) and Engineering Research Building (ERB). The system consists of standalone kiosks placed in the main lobby of ERSA and at the main entrances of the ERB, with

functionality tailored to each building's purpose. In ERB, the kiosk serves as an informational hub, offering an interactive building map and a dynamic announcement board to assist navigation and communication. In ERSA, the system expands to include secure user login with card swipe input option, conference room reservations, and supply requests to support research and faculty operations. The application is also available as a web-based platform for remote access, though certain features are intentionally limited to preserve the kiosk's role as an on-site resource. Tests will be conducted to measure metrics such as user friendliness, maintainability, and overall performance. Through these kiosks, we aim to enhance the overall user experience at UTA by centralizing information and streamlining everyday tasks.

Background

Navigating large campus buildings and accessing essential resources can be inefficient and frustrating for students, faculty, and visitors. UTA is committed to improving accessibility and the overall campus experience for students, faculty, and visitors. However, several challenges exist within key buildings such as ERSA and ERB. At ERSA, faculty reported issues with missing supplies and the lack of a structured system for reserving conference rooms. In the six-story ERB, students and visitors often struggle to locate rooms and events due to the building's size and complexity. To address these challenges, we developed the Smart Kiosk system, a centralized platform that provides interactive navigation, room reservations, and supply requests. This solution improves day-to-day efficiency and enhances the overall campus experience, with the potential to scale across additional buildings.

Project Requirements

1. Room Reservation System for ERSA
2. Supply Request System for ERSA
3. Automated Email System for Requirements (1) & (2)
4. Interactive Map for ERSA & ERB

5. Secure User Login for ERSA
6. Panel for CSE Announcements & Banners
7. Admin Dashboard
8. Website for Remote Access (Mobile Compatible)
9. Reliable Auto Startup after Reboot
10. Kiosk Mode to keep Users on the Kiosk System

Design Constraints

- **Physical Constraints:**

- The screen sizes must be larger than 18 inches with proper screen protection. The display must feature 1080p resolution or better.
- Touch screen latency may vary depending on how CPU intensive the task may be.
- Use of the Kiosk must remain indoors preferably at building entrances.
- The Smart Kiosk must be completely packaged and secure before transportation.
- No food and drinks shall be near the Smart Kiosk upon use.

- **Technical & Software Constraints:**

- The Supply Request system will not specify the amount of supplies that should be delivered or the specific location to where these supplies will be delivered.
- The mapping feature shall not provide step-by-step written directions or numeric distance values.
- Software used for the Smart Kiosk must be compatible with the internal computer (Rpi 5). The local web server shall operate on ports 8000 and 5173. An internet connection must be established for use.
- The Smart Kiosk website utilizes MappedIn for the mapping functionalities, therefore its functionality will depend on them.
- The reservation confirmation speed must be met regardless of the user's operation (reservation creation, deletion and alteration).

- Connection to the website for the Smart Kiosk shall be dependent on its WiFi connection to ERSA's and ERB's building internet router.
- **Maintenance Constraints:**
 - The Smart Kiosk's idle states will require images to be updated frequently by the kiosk administrators.
 - The remote website is hosted on Render on a paid plan, which requires a recurring monthly payment to maintain availability.
 - Maintenance activities will be limited to authorized personnel with access credentials to the GitHub repository and administrative privileges on the kiosk system. Physical maintenance shall depend on the availability of replacement parts and access to the kiosk location. Remote software maintenance will require reliable internet access to the kiosk network.
- **User Access Constraints:**
 - The remote website must require users to be connected to the internet. Using the website requires an account.
 - Users must have their UTA ID card to utilize the card-swipe login feature.
 - The website application must approve the log in attempt if the user provided a matching username (email, pin, etc) and password, and deny if otherwise.
 - Only administrative users shall have permission to modify system settings.
 - The end-users shall not have access to any other computer applications or websites not related to the product.

Engineering Standards

- **Software Standards:**
 - ISO/IEC 14764:2006 - Software Life Cycle Processes - Maintenance
 - ISO/IEC 12207:2017 - Software Life Cycle Processes
 - IEEE 828-2012 - Standard for Configuration Management in Systems and Software Engineering

- IEEE 16326-2009 - Software Life Cycle Processes - Project Management
- Simple Mail Transfer Protocol (SMTP) - Standard Protocol for Email Transmission
- ISO/IEC 25010:2011 - System and Software Quality Models
- ISO/IEC 23026:2015 - Systems and Software Engineering - Website Engineering
- **Usability and Accessibility Standards:**
 - ISO 9241-210:2019 - Ergonomics of Human-System Interaction - Human-Centered Design
 - ISO 9241-110:2020 - Ergonomics of Human-System Interaction - Dialogue Principles
 - ISO 9241-171:2008 - Guidance on Software Accessibility
 - ISO 9241-303:2011 - Requirements for Electronic Visual Displays
 - ISO 13406-2 - Ergonomic Requirements for Flat Panel Displays
 - WCAG 2.1 - Web Content Accessibility Guidelines
 - ADA Title III - Accessibility Requirements for Public Accommodations
 - ISO 8601 - Date and Time Format Standard
 - ANSI/INCITS 322-2008 – Information Technology - Role-Based Access Control (RBAC)
- **Security Standards:**
 - ISO/IEC 27001:2022 - Information Security Management Systems (ISMS) Requirements
 - NIST SP 800-63B - Digital Identity Guidelines - Authentication and Lifecycle Management
 - ISO/IEC 29115:2013 - Entity Authentication Assurance Framework
 - Sensitive information must be transferred securely. Licensing terms of the software used must be properly documented.
- **Safety Standards:**
 - NFPA 70 - National Electrical Code (NEC)
 - 29 CFR 1910.147 - Occupational Safety and Health Standards - Control of Hazardous Energy (Lockout/Tagout)

- IEC 60950-1 / IEC 62368-1 - Safety Standards for Information Technology and Audio/Video Equipment

System Overview

- **Presentation Layer:**

- The Presentation Layer, also referred to as the front-end of the web application, is what the end-user sees and interacts with on the Smart Kiosk. Our team has placed an emphasis on making sure this layer provides an interface that is aesthetically pleasing, user-friendly, and representative of UTA's culture. The front-end achieves this by using libraries and web frameworks such as **React.js**, **JSX**, and **CSS** to create a sleek and navigable graphical user interface that displays information and captures user input. Some components of this layer include the user interfaces for login and registration, room reservation feature, supply request feature, mapping functionality, and the home page. Finally, this layer exclusively communicates with the backend through **APIs** in a client-server relationship to ensure user requests are communicated properly. This system is deployed on **Render**.

- **Backend Layer:**

- The Backend layer of the Smart Kiosk is responsible for managing the business and server-side logic of the web application. This includes some of the core functionalities of the Smart Kiosk such as room reservations, supply requests and the ability to create user profiles. All operations in this layer are performed behind the scenes and are invisible to end-users. The Smart Kiosk's backend is implemented using **Django**, a Python web framework, to efficiently manage user authentication, authorization, and email handling. **REST APIs** will be implemented through Django to allow for communication with the front-end and database layers. Some examples include integrating **Microsoft Outlook** and **MappedIn**. These APIs act as the middleman, and ensure the integrity and security of the

data such as user input and client requests. This system is deployed on **Render**.

- **Database Layer:**

- The Database Layer of the Smart Kiosk manages how the backend stores and retrieves data. This layer utilizes **TiDB**, a **SQL** database, to store information such as user profiles, room reservations, supplies and their requests, among other data. The backend will be able to access this database through **Django Object-Relational Mapper (ORM)**. Django ORM abstracts database information and acts as a translator of Python to SQL queries, allowing for efficient and secure communication between the two layers.

- **Hardware Layer:**

- The Hardware Layer of the Smart Kiosk encompasses all the physical components that end-users can touch and see, and support the operation of the web application. Its functionality is to serve as a platform where visitors of the ERSA building can access the web applications functionalities through a large, sturdy device. The features of this layer include a touchscreen display, a **Raspberry Pi 5** single-board computer, WiFi connectivity, a card swiper reader, and a secure stand. These features enhance the accessibility, computing power, and foundation of the Smart Kiosk.

Results

- Evaluation of the Smart Kiosk is based on a series of usability and performance tests based on real-world user scenarios. Participants included students and faculty members whose feedback was used to improve usability and system reliability.
- **Test cases:**
 - User login - card swipe, manual input
 - Map - Navigation and Identification
 - Conference Room Reservation

- Supply Requests
- **Performance metrics:**
 - Task completion time
 - System response time
 - Error rates during interaction
 - User satisfaction and ease of use

Metric Name	Metric Description	Metric Rating
User Rating	Average faculty and student rating based on their experience interacting with the UI	8.2/10
'Page Speed' Accessibility Rating	Performance measure to highlight the accessibility of our web app performed through an automatic AI service.	80%
Swipe Card Login Time	Time it takes for user to log-in to the system using the swipe card system	3.95 seconds
Dashboard Loading Time	Time it takes for the Raspberry Pi to render the ERSA Smart Kiosk Dashboard	3.10 seconds
RPI vs Mini PC performance	Comparison to see how well different kiosk computers run the smart kiosk website. Raspberry Pi vs. KAMRUI Ryzen 7 Mini PC	Mini PC is about 4x more efficient

Future Work

- Upgrade the Kiosk Computer to a Mini PC to improve latency

Project Files

- Project Charter
- System Requirements Specification
- Architectural Design Specification
- Detailed Design Specification
- Poster
- Closeout Materials

References

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